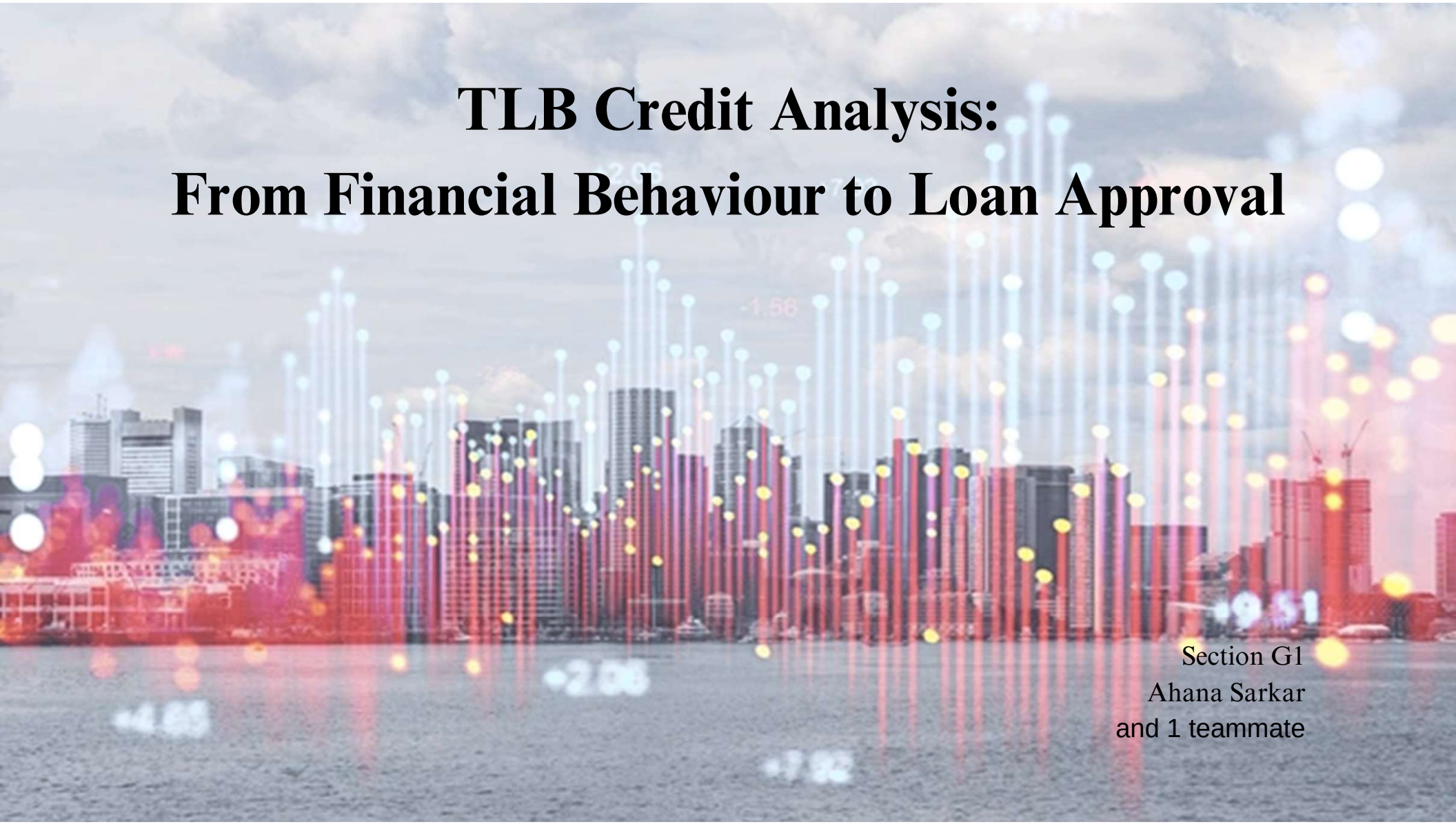


TLB Credit Analysis:
From Financial Behaviour to Loan Approval

Section G1
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and 1 teammate

The Story We Set Out to Tell

Tracing how financial behavior shapes credit ratings and ultimately determines loan approval outcomes

- What is the distribution and relationship between applicant characteristics and Credit Rating, and which factors influence it the most?
- Which financial behaviour factors significantly predict Credit Rating, and does Credit Rating explain the differences in loan approval outcomes?

Key Questions We Investigate

- Financial behaviour is predictable and measurable
- Credit ratings are driven by observable financial signals
- Loan approval outcomes follow credit rating patterns
- Data-driven insights can improve credit evaluation

What the Analysis Shows

- TLB can refine its credit evaluation using data-driven risk factors.
- Approval criteria can be adjusted for misclassified groups.
- Rejected applicants can be guided on how to improve approval chances.
- This creates opportunities for customer retention through financial guidance.

Why This Matters

Our Approach:

- Data Cleaning
- Exploratory Data Analysis
- Inferential Statistics
- Recommendations

Final Takeaway

Our analysis will show that loan approval outcomes are not random but follow predictable financial behaviour patterns.

By leveraging these insights, TLB can improve the accuracy and fairness of its lending decisions while also guiding customers on how to strengthen their financial profiles and increase their chances of approval.

Data Cleaning

1. Changing Incorrect Data Types

Incorrect data types were identified for the columns 'Telephone', 'Foreign_Worker', 'Age', and 'Existing_Credits'. These were converted to integer types, as age and existing credits cannot contain decimal values, and the telephone and foreign worker variables represent binary options (TRUE/FALSE).

For simplicity, 'Instalments' was also kept as an integer, instead of representing percentage values as floats.

2. Adjusting for Missing Values

The variables 'Age' and 'Existing_Credits' contained 56 and 58 missing values, respectively. Since these counts were too large to ignore, the missing values were imputed.

For Age, the median was used to prevent skewness from affecting the replacement values.

For Existing_Credits, the mode was used since the variable has only three possible integer values, making the most frequent value the most representative.

3. Removing Trailing Spaces

Trailing spaces were removed across all variables to ensure data consistency and accurate matching, preventing potential analysis errors or skewed results.

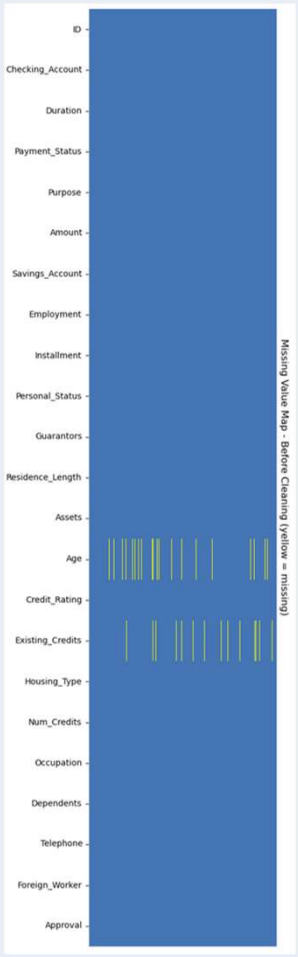
4. Removing Incomplete Variable Inputs

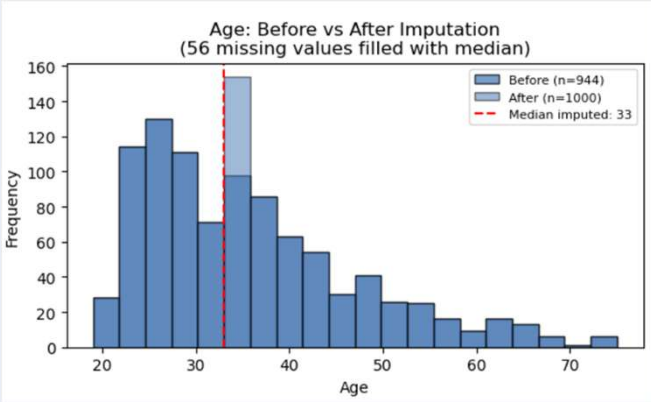
The variable 'Personal_Status' contained multiple relationship categories for men but only a single category for women, with no differentiation in relationship status. Because this variable lacked consistent information across genders, it was removed from the analysis.

5. Savings Account Data Limitation

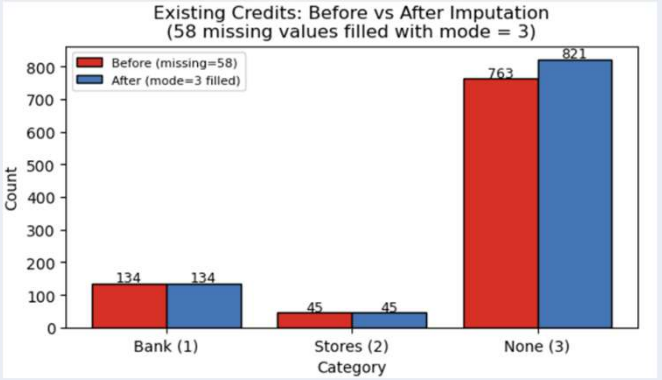
The Savings Account variable is recorded in broad brackets, with no granularity for values above $1,000. As a result, customers with $1,200 and $500,000 would be treated the same.

To avoid false precision, savings values above $1,000 were grouped into a single bracket and treated as categorical rather than continuous data in the analysis.



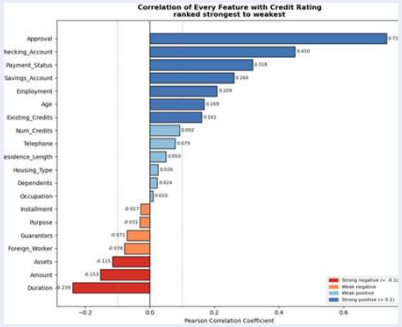


After the data cleaning applications, the graphs show the changes made to age; the median age rises in counts, as that was chosen for the missing variables. As for existing credits, we see that the only real change occurs in the mode, keeping the other categories the same as before the cleaning.



EDA: What Drives Credit Rating

To understand the distribution and relationships between applicant characteristics (age, employment, assets, savings, payment history, checking account, etc.) and credit rating, we conducted an exploratory data analysis (EDA) to identify which factors influence creditworthiness the most.

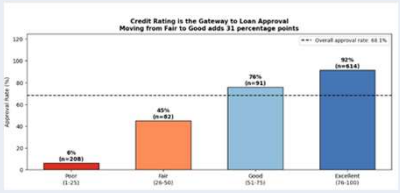


Key EDA findings:

The analysis shows that credit rating plays a central role in loan approval decisions. Customers with higher credit ratings have a significantly greater probability of having their loans approved.

Financial behaviour indicators such as checking account balance, payment history, savings level, and employment stability are strongly associated with credit rating. A stronger credit rating increases the likelihood of loan approval.

We also observed that loans that are approved tend to involve smaller loan amounts and shorter durations, suggesting that borrowers requesting lower amounts over shorter repayment periods are perceived as lower risk.



Important Anomalies Identified

Several interesting patterns emerged during the EDA that challenge initial assumptions:

Credit Rating of -1:

This single record likely represents a data entry error or system placeholder. The applicant (age 23) was rejected despite otherwise meeting typical loan requirements.

Critical Account Holders Have Higher Credit Ratings Than "All Paid":

52% of critical account holders have a checking account balance above $200, compared with only 17.5% of applicants with no credit history. This suggests that "critical" accounts may represent high-activity borrowers rather than defaulters, while having no credit history is penalised more strongly.

No Checking Account + High Savings Still Achieves Decent Credit Ratings:

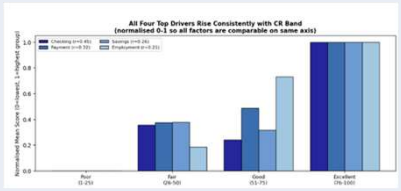
Savings can partially compensate for not having a checking account. Applicants with savings above $500 still achieve relatively strong credit ratings (CR ≈ 70.8).

Loan Duration × Amount Interaction:

Loans with both high amounts and long durations tend to have noticeably lower credit ratings and approval probabilities.

No Existing Credits = Higher Credit Rating Than Bank Credits:

Applicants with no other credits (CR ≈ 73.1) outperform those with active bank credits (CR ≈ 57.7). Additional credit obligations increase perceived risk.



Co-Applicant vs Guarantor:

Co-applicants show lower approval rates (≈56%) than both guarantors (≈75%) and applicants without guarantors (≈68%). A guarantor signals external confidence, whereas a co-applicant may indicate the applicant cannot qualify independently.

Foreign Worker Effect:

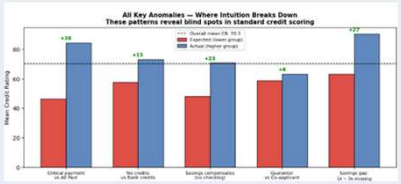
Non-foreign workers show higher credit ratings (CR ≈ 84.6 vs 69.8). However, this group contains only 37 observations and may reflect sample bias rather than a true effect.

Optimal Number of Existing Credits:

Credit ratings peak at three existing credits (CR ≈ 81.4) before declining when the number of obligations increases further.

High-Amount Loans with Poor Credit Ratings Still Approved:

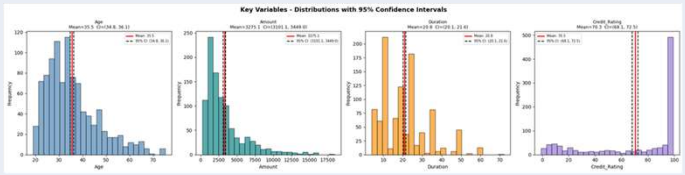
Approximately 17% of high-amount applicants with poor credit ratings are still approved, often when a guarantor is present, indicating that third-party security can offset risk.



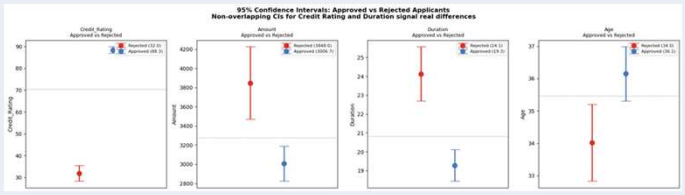
Are the patterns observed real, or coincidence ?

In this inferential statistical analysis, we checked the relationship between applicant financial behaviour and loan approval mediated by Credit Rating, and which behavioural factors are statistically significant predictors of that score.

The correlation analysis returned a correlation of 0.733 between credit rating and approval.
Before testing hypotheses, we need to know how well our sample of 1000 represents the true population of TLB applicants. A 95% confidence interval here means: if we repeated this study 100 times with different samples, 95 of those intervals would contain the true population mean.
We used the t-distribution because the population standard deviation is unknown. The variables selected are the four continuous features most relevant to the problem statement: Age, Amount, Duration, and Credit Rating.



Credit Rating CIs do not overlap at all, this is a strong visual signal before any formal test is run that the difference is almost certainly real

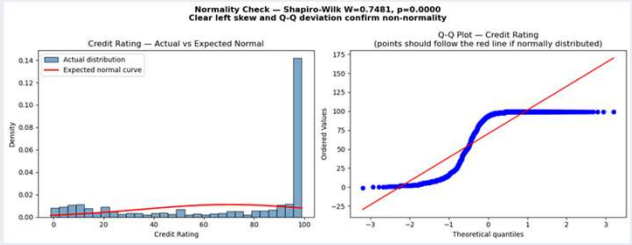


Normality Check
This is the decision point for the entire hypothesis testing section.
Parametric tests rely on two core assumptions:

1. The dependent variable is approximately normally distributed within each group
2. Groups have roughly equal variance (homoscedasticity)

We tested both assumptions before choosing our test method. If either failed, we cannot trust the p-values from parametric tests, for this could indicate false positives or overly conservative values.

Why this matters for our data specifically: Credit Rating is left-skewed, meaning that this shape already suggests normality will fail before we even run the test.
The results gave us $p < 0.05$, so we reject normality assumptions. Credit Rating is thus not normally distributed. This means ANOVA and t-test assumptions are violated.

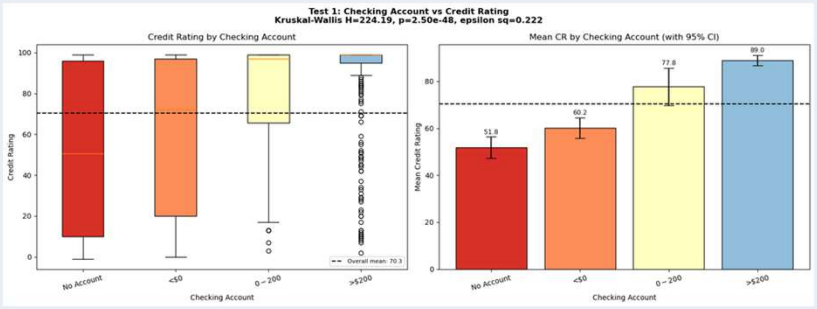


Levene's test for equal variance rejected the equal variance assumption, because $p < 0.05$, so an ANOVA test would not have been appropriate.

Hypothesis Testing

Hypothesis 1: Applicants with higher checking balances have higher credit ratings. The No Account group had a mean CR of 51.8 vs 89.0 for the >$200 group.

H0: The distribution of Credit Rating is the same across all Checking Account groups
H1: At least one Checking Account group has a significantly different Credit Rating distribution
Why Kruskal-Wallis: Credit Rating failed normality (Shapiro-Wilk $p < 0.001$) and equal variance (Levene $p < 0.05$). Kruskal-Wallis is the correct non-parametric replacement for one-way ANOVA as it compares rank distributions rather than means, making no assumption about normality or variance.
alpha = 0.05



Hypothesis Results:

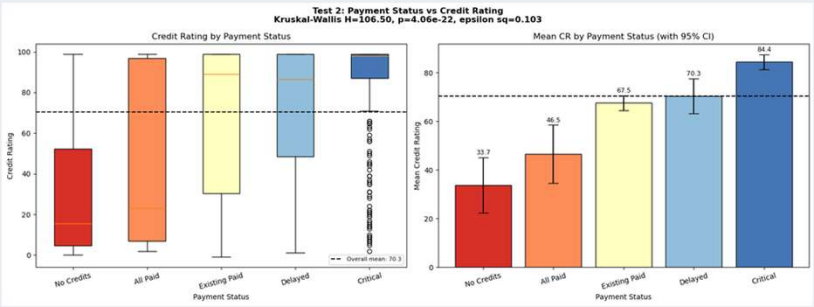
We reject the null hypothesis

Checking Account is the single strongest categorical predictor of CR. A 37-point gap between the lowest and highest groups, with 5 of 6 pairwise pairs significant.

Hypothesis Test 2: Applicants with better payment records have higher credit ratings. The EDA also flagged the critical account paradox: Critical (CR=84.4) > All Paid (CR=46.5).

H0: The distribution of Credit Rating is the same across all Payment Status groups
H1: At least one Payment Status group has a significantly different Credit Rating distribution

Why Kruskal-Wallis: Credit Rating failed normality (Shapiro-Wilk $p < 0.001$) and equal variance (Levene $p < 0.05$). Kruskal-Wallis is the correct non-parametric replacement for one-way ANOVA, it compares rank distributions rather than means, making no assumption about normality or variance.
alpha = 0.05



Hypothesis Results:

We reject the null hypothesis

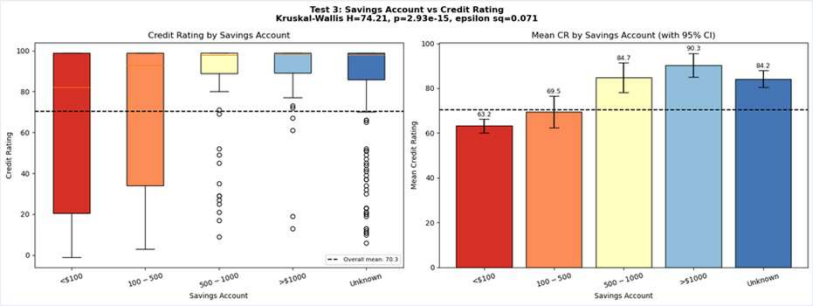
Payment Status significantly predicts CR and the critical account paradox holds.
No borrowing history is penalised more than a critical account because the bank has zero evidence of repayment behaviour.

Hypothesis Testing

Hypothesis 3: Applicants with higher savings balances have higher credit ratings.

H0: The distribution of Credit Rating is the same across all Savings Account groups
H1: At least one Savings Account group has a significantly different Credit Rating distribution

Why Kruskal-Wallis: Credit Rating failed normality (Shapiro-Wilk $p < 0.001$) and equal variance (Levene $p < 0.05$). Kruskal-Wallis is the correct non-parametric replacement for one-way ANOVA it compares rank distributions rather than means, making no assumption about normality or variance.
alpha = 0.05



Hypothesis Results:

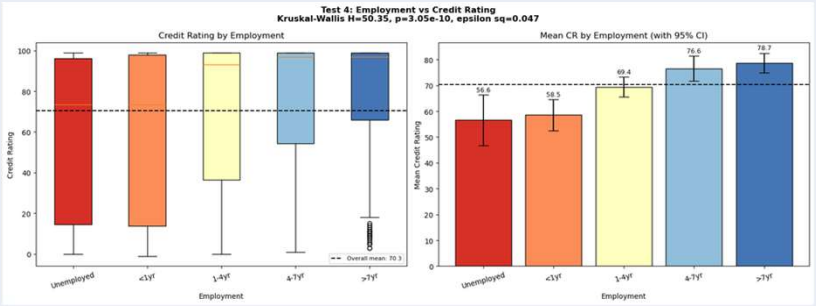
We reject the null hypothesis

Savings Account significantly predicts CR. The >$1000 savings group achieves mean CR = 90.3 vs 63.2 for <$100. The EDA compensation anomaly is statistically backed.

Hypothesis Test 4: Longer employment tenure predicts higher credit ratings. EDA cross-factor finding: older unemployed applicants (likely retired) score higher than expected due to savings, not employment.

H0: The distribution of Credit Rating is the same across all Employment groups
H1: At least one Employment group has a significantly different Credit Rating distribution

Why Kruskal-Wallis: Credit Rating failed normality (Shapiro-Wilk $p < 0.001$) and equal variance (Levene $p < 0.05$). Kruskal-Wallis is the correct non-parametric replacement for one-way ANOVA, it compares rank distributions rather than means, making no assumption about normality or variance.
alpha = 0.05



Hypothesis Results:

We reject the null hypothesis

Employment significantly predicts CR. >7yr employment mean CR=78.7 vs Unemployed CR=56.6. EDA retired paradox holds: the effect is moderated by age and savings.

Hypothesis Testing

Hypothesis 5: Approved applicants take smaller, shorter loans. Duration (r=-0.24 with CR) and Amount (r=-0.15 with CR) are the two strongest negative correlates in the dataset. The EDA heatmap showed that these penalties compound, a bigger AND longer loan pushes CR significantly lower than either alone.
Why this test uses Approval as the grouping variable and not CR is because tests 1-4 proved financial behaviour predicts CR.

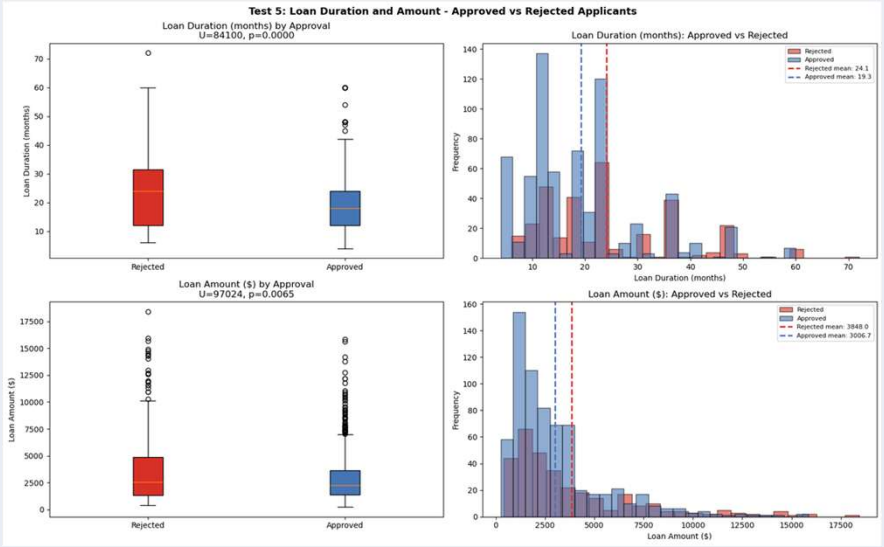
This test asks the next question: do the loan characteristics themselves differ significantly between those the bank approved and those it rejected? Here Approval is used as a binary grouping variable to split the sample, and Duration and Amount are the continuous outcomes being compared. This is the right arrow in our causal chain: CR -> Approval.

H0: Loan duration is the same for approved and rejected applicants H1: Loan duration differs between approved and rejected applicants
H0: Loan amount is the same for approved and rejected applicants H1: Loan amount differs between approved and rejected applicants

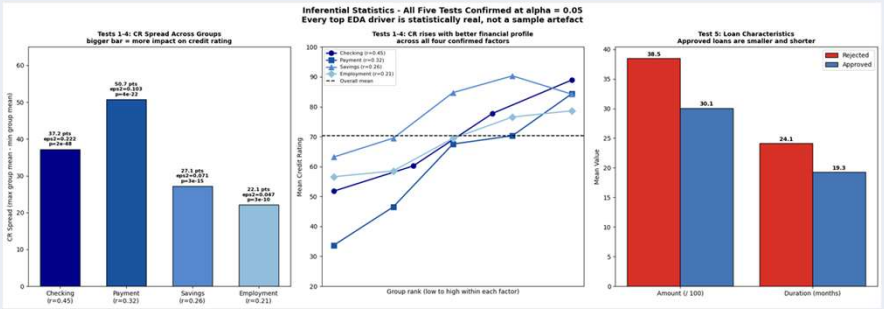
Why Mann-Whitney U: two-group comparison on a continuous variable where normality cannot be assumed. Mann-Whitney U is the non-parametric replacement for an independent t-test, it compares rank distributions rather than means.
Spearman correlations are also reported to show the continuous relationship between Duration/Amount and Credit Rating directly (not just via the approval split).
alpha = 0.05

Hypothesis Results:
Rejected applicants ask for more money over a longer period.
Reducing loan amount or term can shift a borderline case into approval.

Duration: approved=19.3m vs rejected=24.1m
Amount: approved=$3007 vs rejected=$3848



Overall test summary



Recommendations

The EDA told a story, inferential statistics proved it, so how do we proceed ?

Strategic Takeaway:

Creditworthiness is driven by observable financial behaviour. By focusing on checking account status, repayment history, savings, employment stability, and loan structure, TLB can improve approval accuracy while giving applicants clearer pathways to qualify for credit.

Checking Account Access	Build Credit History	Savings as Alternative Signal	Employment Stability Signal	Loan Size & Duration Adjustment
Actionable: opening and funding a checking account is the most direct single step a rejected customer can take to raise their credit rating.	Actionable: customers with no credit history should open a small credit line and repay it consistently before applying for a larger loan.	Actionable: savings is a viable alternative pathway for customers who cannot immediately open a checking account.	Actionable: employment stability is a meaningful signal but should be assessed alongside age and savings, not in isolation.	Actionable: loan size and duration should be adjusted for borderline applicants. Smaller loan amounts or shorter repayment periods can significantly improve approval likelihood.

For TLB: The five confirmed factors: checking account status, payment history, savings level, employment stability, and loan characteristics; provide TLB with a data-backed foundation to refine its credit scoring model.

The bank can adjust risk weights for misclassified groups (such as critical account holders and applicants with no credit history) and replace outright rejections with structured counter-offers on loan size or repayment terms.

For the customer: Applicants can improve their approval chances by opening a checking account, building savings above $500, maintaining a consistent repayment record, and applying for smaller or shorter-term loans.

These steps are no longer intuitive suggestions, they are supported by statistically validated evidence from the analysis.